DEEP LEARNING MODULE

Workshop 3 : Designing a feed forward network

This workshop aims to give you experience at developing a feed forward network in Keras.

1. Download the code for the MNIST example, run it and review it to make sure you understand the code.

2. The Fashion MNIST data set is a similar data set. It has 10 categories of fashion items (T-shirts, Trousers, … Bag, Ankle Boots), 60,000 training and 10,000 testing examples. Like MNIST, each example consists of a 28x28 grayscale image and its category.
    a. To load the data, use:

```
from keras.datasets import fashion_mnist

(x_train, y_train), (x_test, y_test) = fashion_mnist.load_data()
```

    b. Design a feed forward neural network for this data, using the steps we used for the MNIST example we covered in the lecture as a guide.
    c. Use the "summary" method to see the shapes and the number of parameters. E.g. network.summary()
    d. Experiment with different number of layers, types of activation functions, epochs to see their effect.

    Note, that you can experiment with different learning rates by specifying an argument to RMSprop. For example, the following specifies a learning rate(lr) of 0.005.

```
From keras import optimizers
network.compile(optimizer=optimizers.RMSprop(lr=0.005),
loss='categorical_crossentropy', metrics = ['accuracy'])
```